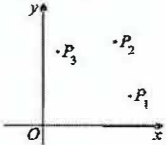
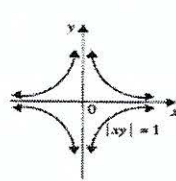
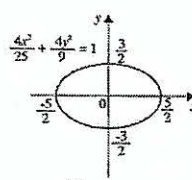
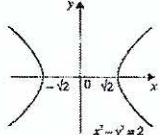
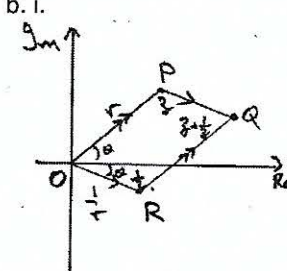


Complex Number 7

- Indicate on an Argand diagram the locus of the point P representing z , given that $|z^2 - \overline{z^2}| = 4$.
- The complex number z is given by $z = t + \frac{1}{t}$, where $t = r(\cos \theta + i \sin \theta)$. Find the equation of the locus of the point P which represents z on an Argand diagram in each of the following cases
 - $r = 2$ and θ varies
 - $\theta = \frac{\pi}{4}$ and r varies
- Solve
 - $x^2 + 2x + 2 = 0$
 - $x^2 + (2 - i)x - 2i = 0$
- In the diagram, the points P_1, P_2 and P_3 represent the complex number z_1, z_2 and z_3 respectively. If $\frac{z_2}{z_1} = \frac{z_3}{z_2}$, show that OP_2 bisect $\angle P_1OP_3$.
 
- If z_1 and z_2 are complex numbers such that $|z_1| = |z_2|$, show that: $\arg(z_1 z_2) = \arg((z_1 + z_2)^2)$.
- Let $z_1 = 2i$ and $z_2 = 1 + \sqrt{3}i$.
 - Express z_1 and z_2 in mod-arg form.
 - Plot in the complex plane the points P, Q, R and S representing $z_1, z_2, z_1 + z_2$ and $z_1 - z_2$ respectively.
 - Find the exact values of: i. $\arg(z_1 + z_2)$ ii. $\arg(z_1 - z_2)$
- Find the values of z such that $z^2 = 5 - 12i$.
 - Solve the equation $z^2 + (4i - 7)z + 7 - 11i = 0$
 - Solve and describe the solutions of: $|z^2 + (4i - 7)z + 7 - 11i| = |z - 2 + i|$
- A polynomial $P(z)$ is defined by: $P(z) = z^6 + 4z^4 + 16z^2 + 64$.
 - Find $P(2i)$
 - Deduce a quadratic factor of $P(z)$ giving reasons.
 - Factorise the polynomial $P(z)$ over the set of real number $\mathbb{R}$.
 - Factorise the polynomial $P(z)$ over the set of complex number $\mathbb{C}$.
- A Polynomial $P(x)$ has all real coefficients and z is a complex number. If $P(z)P(\bar{z}) = 16$ find the value of $|P(z)|$ giving reasons.
- Show that $\frac{(1+i)^8}{(1-\sqrt{3}i)^k} = 2^{4-k} \left[\cos\left(\frac{k\pi}{3}\right) + i \sin\left(\frac{k\pi}{3}\right) \right]$
 - For what values of k is $\frac{(1+i)^8}{(1-\sqrt{3}i)^k}$ purely imaginary?
- If w is one of the complex cube roots of unity, prove that the other complex root is w^2
 - Show that $1 + w + w^2 = 0$
 - Prove that if n is a positive integer, then $1 + w^n + w^{2n} = 3$ or 0 , depending on whether n is or is not a multiple of 3.
- Use De Moivre's Theorem to show that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$ where n is an integer ($\cos \theta \neq 0$).
 - Use this result to show that the equation $(1 + z)^4 + (1 - z)^4 = 0$ has roots of $\pm i \tan \frac{\pi}{8}, \pm i \tan \frac{3\pi}{8}$
 - Hence or otherwise, show that $\tan^2 \frac{\pi}{8} = 3 - 2\sqrt{2}$
- Given that ω is one of the non-real roots of $z^3 = 1$ show that $1 + \omega + \omega^2 = 0$
 - Using part a., or otherwise, show that $\left(\frac{\omega}{1+\omega}\right)^k + \left(\frac{\omega^2}{1+\omega^2}\right)^k = (-1)^k 2 \cos \frac{2}{3} k\pi$ where $k \in \mathbb{Z}$.

14. a. Solve $z^9 - 1 = 0$ and determine the six solutions of $z^6 + z^3 + 1 = 0$
 b. Hence show that $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9}$
15. Sketch on the argand diagram the points z which satisfy $|z - 1| \leq |z + i|$
16. a. Show that if $\operatorname{Re} \left[\frac{z-1}{z+1} \right] = 0$ then z lies on the circle $x^2 + y^2 = 1$.
 b. Find z such that $\operatorname{Re} \left[\frac{z-1}{z+1} \right] = 0$ and $\arg(z + 1) = \frac{\pi}{6}$.
17. If ω is a non-real cube root of unity show that $(1 + \omega^2)(\omega^{-1} + \omega^2) = -2$
18. a. Find, in term form 'cis θ ', the roots of the equation: $z^{2n+1} = 1$, where $n = 0, 1, 2, \dots$
 b. Hence factorise $z^{2n} + z^{2n-1} + \dots + z + 1$ into quadratic factors with real coefficients.
 c. Hence, or otherwise find $2^n \times \sin \frac{\pi}{2n+1} \times \sin \frac{2\pi}{2n+1} \times \dots \times \sin \frac{n\pi}{2n+1}$
19. a. Express the roots of the equation $z^5 + 32 = 0$ in modulus/argument form.
 b. Hence show that $z^4 - 2z^3 + 4z^2 - 8z + 16 = (z^2 - 4 \cos \frac{\pi}{5} z + 4)(z^2 - 4 \cos \frac{3\pi}{5} z + 4)$
 c. Hence find the exact value of $\cos \frac{\pi}{5}$ in simplest surd form
20. a. Given that $z = r \operatorname{cis} \theta$ express $\frac{1}{z}$ in terms of r and θ .
 b. The origin O and the points P, Q and R , representing the complex number $z, z + \frac{1}{z}$ and $\frac{1}{z}$ respectively, form a quadrilateral. Assume that $0 < \arg z < \frac{\pi}{2}$ and $|z| \geq 1$
 i. Sketch the quadrilateral $OPQR$ on the complex plane.
 ii. Find the complex number z for $OPQR$ to be a square.
21. Three points A_1, A_2, A_3 representing the complex numbers z_1, z_2, z_3 in the Argand diagram are equally spaced around the circumference of the circle $|z| = 1$.
 a. Show that $z_1 + z_2 + z_3 = 0$.
 b. The point P represents the complex number ω and $|\omega| = 3$. Find, by using the fact that $z\bar{z} = |z|^2$, or otherwise, the value of $|PA_1|^2 + |PA_2|^2 + |PA_3|^2$.

Answer:

1. 
2. 

3. $x = -1 \pm i; x = -2, i$
6. a. $z_1 = 2 \operatorname{cis} \frac{\pi}{2}, z_2 = 2 \operatorname{cis} \frac{\pi}{3}$ c. i. $\frac{5\pi}{12}$
 ii. $\frac{11\pi}{12}$
7. a. $z = \pm(3 - 2i)$ b. $z = 5 - 3i$ or $2 - i$. c. $z = 2 - i$, the point $(2, -1)$, or $|z - (5 - 3i)| = 1$, the circle $(5, -3)$, radius 1 unit.
8. a. $P(2i) = 0$ b. $(z - 2i)(z + 2i) = (z^2 + 4)$ is a real quadratic factor.
 c. $P(z) = (z^2 + 4)(z^2 - 2\sqrt{2}z + 4)(z^2 + 2\sqrt{2}z + 4)$. d. $P(z) = (z + 2i)(z - 2i)(z - \sqrt{2} - \sqrt{2}i)(z - \sqrt{2} + \sqrt{2}i)(z + \sqrt{2} - \sqrt{2}i)(z + \sqrt{2} + \sqrt{2}i)$
9. 4
10. b. $k = \frac{3}{2} + 3n$ (n is integer)
14. b. For $z^9 - 1 = 0, z = \operatorname{cis} \frac{2k\pi}{9}, k = 0, 1, 2, \dots, 8$. For $z^6 + z^3 + 1 = 0, z = \operatorname{cis} \pm \frac{2\pi}{9}, \operatorname{cis} \pm \frac{4\pi}{9}, \operatorname{cis} \pm \frac{8\pi}{9}$
15. $y \geq -x$
16. b. $z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$
18. a. $z_n = 1 \operatorname{cis} \frac{2k\pi}{2n+1}, k = \pm 1, \pm 2, \dots, n$, for $n = 0, 1, 2, \dots$; b. $\sum_{m=0}^{2n} z^m = \prod_{k=1}^n [z^2 - 2z \cos \frac{2k\pi}{2n+1} + 1]; c. \sqrt{2n+1}$
19. a. $z_1 = 2 \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right), z_2 = 2 \left(\cos \left(\frac{3\pi}{5} \right) + i \sin \left(\frac{3\pi}{5} \right) \right), z_3 = 2(\cos \pi + i \sin \pi) = -2, z_4 = 2 \left(\cos \left(\frac{7\pi}{5} \right) + i \sin \left(\frac{7\pi}{5} \right) \right) = 2 \operatorname{cis} \left(-\frac{3\pi}{5} \right), z_5 = 2 \left(\cos \left(\frac{9\pi}{5} \right) + i \sin \left(\frac{9\pi}{5} \right) \right) = 2 \operatorname{cis} \left(-\frac{\pi}{5} \right)$; c. $\cos \frac{\pi}{5} = \frac{1+\sqrt{5}}{4}$
20. a. $\frac{1}{z} = \frac{1}{r} (\cos(-\theta) + i \sin(-\theta))$; or $\frac{1}{r} \operatorname{cis}(-\theta)$
 b. i. 
 ii. $z = \operatorname{cis} \frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$
21. b. 30